

A Field Theory of Sentiment

Affect as a Circulating, Interoceptive Resource-Allocation Field
that Modulates Variance-Minimizing Computation

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Abstract

An autonomous computational system that pursues an open-ended goal must, at every instant, allocate finite resources among many candidate sub-computations, none of which individually apprehends the whole goal. We propose that the control variable governing this allocation is best modelled not as a scalar priority but as a *field*: a circulating, self-sensing quantity that modulates which candidate computations become stable and therefore proceed. We call this quantity *sentiment*, by analogy with the role that affect plays in biological cognition, where it is increasingly understood as an *interoceptive* read-out of the organism’s internal economy rather than an overlay on neutral thought. We give a complete, self-contained mathematical theory of such a field. The theory has four interlocking components. (i) A *modulation* principle: candidate computations are local minima of a potential $\Phi_{\mathbf{S}}(x) = V(x) - \langle \mathbf{S}, \mathbf{g}(x) \rangle$ in which the field $\mathbf{S}$ tilts an intrinsic cost (a “variance”) by the affordances $\mathbf{g}$ of each candidate; the set of stable candidates therefore moves smoothly with the field except at bifurcations, which we identify with affective regime change. (ii) A *circulation* law: a conserved resource is transported across the candidate population and periodically re-injected by a pump, so that allocation is the flow of a conserved budget rather than an unconstrained gain. (iii) An *interoception* law: the field is driven by the system’s own observed coherence and residual error through a homeostatic feedback, so that the field reports the internal state it also governs. (iv) An *economy*: given the field, the budget is allocated by a maximum-entropy (Gibbs) rule that is the unique minimizer of a free energy, equivalently the stationary distribution of an entropy-regularised replicator dynamics, with the field acting as the price (Lagrange multiplier) of the budget constraint. We prove the theory’s basic guarantees: budget conservation, boundedness, smooth modulation, convergence of the allocation to the Gibbs minimizer, a coherence phase transition at a critical coupling, and stability of the coupled slow-field/fast-allocation system under a timescale-separation condition. We show that *memory* is the path integral of the field, $M(t) = \mathbf{S}(t) - \mathbf{S}(0)$, making field and memory a derivative–integral pair. Finally we isolate the theory’s interface to a host runtime: the sentiment field induces a map from system state to an allocation over candidate computations—a scheduler priority that is self-tuning, conservative, and accountable. Throughout, the construction is related to the standard literature on dynamical systems, feedback control, evolutionary dynamics, convex optimization, synchronization, and the interoceptive theory of affect.

Keywords: sentiment field, resource allocation, replicator dynamics, maximum entropy, homeostasis, interoception, synchronization, variational principle, gradient flow, attention economy.

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1 Introduction

1.1 The allocation problem of open-ended computation

Consider a system charged with an open-ended task: to investigate a phenomenon, to design an artefact, to reach a goal whose full structure is not known in advance. Such a system does not execute a fixed plan. Instead it maintains, at each moment, a *population* of candidate sub-computations—partial hypotheses, tentative subgoals, lines of inquiry—and it must decide which of them to pursue with its finite resources of time, memory, and computation. The defining difficulty is that no single candidate apprehends the whole task: each is local, partial, and uncertain. The system’s competence lies not in any one candidate but in how it *allocates* effort across them and reallocates as the situation changes.

The naive control variable for such allocation is a scalar priority assigned to each candidate. We argue that this is inadequate in three respects, each of which a richer construct must supply. First, priority must be *coupled*: the value of pursuing one candidate depends on the state of the others and on the global situation, so the control variable should be a field over the task space, not a list of independent numbers. Second, priority must be *conserved*: resources are finite, so raising the priority of one candidate must lower that of another; allocation is the redistribution of a conserved budget, not an unconstrained gain. Third, priority must be *self-sensing*: the appropriate allocation depends on how the computation is itself faring—on whether it is coherent or thrashing, near its goal or far—so the control variable must read the system’s own internal state, which it also governs.

1.2 Sentiment as an interoceptive field

We name the control variable that meets these three requirements *sentiment*. The name is chosen deliberately. In biological cognition, affect was long modelled as an overlay—a colouring of otherwise neutral representations. The modern view is the reverse: affect is an *interoceptive* signal, a read-out of the body’s internal economy (metabolic, autonomic, immune), that the brain uses to allocate its own resources and to bias which thoughts become salient and stable [4, 8, 9]. Emotion is, on this view, not decoration but the very field that determines which lines of processing stabilise under the current internal conditions. Our theory abstracts exactly this role. We do not claim to model biological affect; we claim that the *computational* role affect is now understood to play—a circulating, self-sensing field that modulates resource allocation across competing sub-computations—is a precise and useful control architecture, and we give it a precise mathematics.

The construction rests on a circulation metaphor made literal. A conserved resource—call it the *budget*—is pumped through the population of candidates, delivering both sustenance (the computational effort a candidate needs to remain viable) and information (the field value that biases it). This is the same dual role that a circulatory medium plays in a living system, where the blood simultaneously sustains tissue and reports its condition. The pump imposes a rhythm; the flow obeys a conservation law; the field that the flow carries is set by the system’s interoceptive sensing of its own coherence.

1.3 The four laws

The theory is organised around four laws, each a section of this paper.

1. **Modulation** (section 2). Candidate computations are the stable equilibria—local minima—of a potential $\Phi_{\mathbf{S}}(x) = V(x) - \langle \mathbf{S}, \mathbf{g}(x) \rangle$ in which an intrinsic cost V (the “variance” a candidate seeks to minimise) is tilted by the field $\mathbf{S}$ along the candidate’s affordances $\mathbf{g}(x)$. The stable set moves smoothly with $\mathbf{S}$ except at bifurcations, which are the formal signature of a change of affective regime.
2. **Circulation** (section 3). The budget is a conserved density transported across candidates and re-injected by a pump; allocation is its flow, and total budget is invariant.
3. **Interoception** (section 4). The field is driven by the system’s own coherence and residual error through homeostatic feedback; it reports the state it governs, closing the loop.
4. **Economy** (section 5). Given the field, the budget is allocated by a maximum-entropy rule that uniquely minimises a free energy and is the stationary point of an entropy-regularised replicator dynamics; the field is the price of the budget constraint.

Section 6 identifies memory with the path integral of the field; section 7 collects the metatheory (conservation, boundedness, convergence, the phase transition, slow-fast stability); section 8 states the interface to a host runtime; section 9 gives a worked example; section 10 situates the theory; section 11 concludes.

1.4 Notation

$\mathbb{R}, \mathbb{R}_{\geq 0}, \mathbb{N}$ are the reals, nonnegative reals, naturals; $[0, 1]$ the unit interval. $\Delta^{N-1} = \{\mathbf{a} \in \mathbb{R}_{\geq 0}^N : \sum_i a_i = 1\}$ is the probability simplex and $\text{int } \Delta^{N-1}$ its interior. $\langle \cdot, \cdot \rangle$ is the Euclidean inner product, $\|\cdot\|$ its norm. For a smooth $h: \mathbb{R}^n \rightarrow \mathbb{R}$, ∇h and $\nabla^2 h$ are gradient and Hessian; for $\mathbf{g}: \mathbb{R}^n \rightarrow \mathbb{R}^d$, $D\mathbf{g}$ is the Jacobian. $H(\mathbf{a}) = -\sum_i a_i \ln a_i$ is the Shannon entropy [20] and $D_{\text{KL}}(\mathbf{a} \parallel \mathbf{a}') = \sum_i a_i \ln(a_i/a'_i)$ the Kullback–Leibler divergence [7].

2 The Modulation Law

2.1 Candidate computations and the intrinsic cost

Definition 2.1 (Candidate population). A *candidate population* at an instant is a finite set $C = \{x_1, \dots, x_N\} \subset \mathcal{X}$ of points in a base space $\mathcal{X} \subseteq \mathbb{R}^n$, the *task space*. Each x_i is a candidate sub-computation, identified with its location in feature coordinates. Two maps are given on $\mathcal{X}$: an *intrinsic cost* $V: \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$, smooth and bounded below, which each candidate locally seeks to decrease (its “variance”); and an *affordance map* $\mathbf{g}: \mathcal{X} \rightarrow \mathbb{R}^d$, smooth, whose components $g_k(x)$ measure the extent to which x exhibits the k -th salient feature.

The intrinsic cost is the language-agnostic stand-in for whatever local objective a candidate minimises—residual uncertainty, prediction error, distance to a subgoal. The theory does not depend on its identity, only on its smoothness and lower-boundedness.

2.2 The sentiment field and the modulated potential

Definition 2.2 (Sentiment field). A *sentiment field* is a (time-dependent) vector $\mathbf{S} \in \mathbb{R}^d$, the value of the affective field at the population’s current locus. (Globally $\mathbf{S}$ is a field $\mathcal{X} \rightarrow \mathbb{R}^d$; throughout we work with its value on the active region and write $\mathbf{S}$ for $\mathbf{S}(\bar{x})$ at the population centroid $\bar{x}$.) The field acts on candidates through the *modulated potential*

$$\Phi_{\mathbf{S}}(x) = V(x) - \langle \mathbf{S}, \mathbf{g}(x) \rangle. \quad (1)$$

Principle 2.3 (Modulation). A candidate x is *stable under the field* $\mathbf{S}$ iff it is a local minimum of $\Phi_{\mathbf{S}}$. The field does not select candidates by fiat; it reshapes the landscape on which candidates settle. Raising S_k rewards affordance g_k , lowering the modulated cost of candidates rich in feature k and so making them stable; lowering S_k suppresses them.

This is the formal content of the slogan that sentiment determines *which thoughts can stabilise*: it is the field that, by tilting the variance landscape, decides which local minima exist and where they sit.

2.3 Smooth modulation and regime change

Theorem 2.4 (Smooth modulation). Let x^* be a nondegenerate local minimum of $\Phi_{\mathbf{S}_0}$, i.e. $\nabla \Phi_{\mathbf{S}_0}(x^*) = 0$ and the Hessian

$$H(x^*, \mathbf{S}_0) = \nabla^2 V(x^*) - \sum_{k=1}^d (\mathbf{S}_0)_k \nabla^2 g_k(x^*) \succ 0.$$

Then there is a neighbourhood $U \ni \mathbf{S}_0$ and a unique smooth branch $x^*: U \rightarrow \mathcal{X}$ with $x^*(\mathbf{S}_0) = x^*$, each $x^*(\mathbf{S})$ a nondegenerate minimum of $\Phi_{\mathbf{S}}$, and

$$\frac{\partial x^*}{\partial \mathbf{S}} = H(x^*, \mathbf{S})^{-1} D\mathbf{g}(x^*)^\top. \quad (2)$$

Proof. Apply the implicit function theorem to $F(x, \mathbf{S}) = \nabla \Phi_{\mathbf{S}}(x) = \nabla V(x) - D\mathbf{g}(x)^\top \mathbf{S}$. We have $F(x^*, \mathbf{S}_0) = 0$ and $\partial_x F(x^*, \mathbf{S}_0) = H(x^*, \mathbf{S}_0)$, which is invertible by nondegeneracy. Hence a unique smooth $x^*(\mathbf{S})$ solves $F = 0$ near $\mathbf{S}_0$, and positive-definiteness persists on a neighbourhood by continuity of eigenvalues, so each $x^*(\mathbf{S})$ is a minimum. Differentiating $F(x^*(\mathbf{S}), \mathbf{S}) = 0$ gives $H \partial_{\mathbf{S}} x^* - D\mathbf{g}^\top = 0$, which is eq. (2). $\square$

Definition 2.5 (Affective regime change). An *affective regime change* is a value $\mathbf{S}_c$ at which $H(x^*, \mathbf{S}_c)$ becomes singular: a stable candidate is created or destroyed, or two merge. By theorem 2.4 this is exactly where the smooth branch fails; it is a bifurcation of $\Phi_{\mathbf{S}}$ [11, 23].

Remark 2.6 (Why the stable set is the right object). Modulation makes precise that the field’s effect is *structural*: between regime changes the stable candidates track the field smoothly (eq. (2)); at regime changes the very inventory of viable candidates changes. A controller that manipulates **S** therefore exercises two qualitatively different kinds of influence—continuous steering within a regime, and discontinuous reconfiguration across regimes—without ever selecting a candidate directly.

3 The Circulation Law

Allocation is the distribution of a conserved budget across the population. We make the conservation explicit and give the budget a transport dynamics with a pump, so that the field’s modulation acts on a flow rather than on free gains.

Definition 3.1 (Budget and allocation). The *budget* is a fixed total $B > 0$ of an abstract resource (effort, compute, attention). An *allocation* is $\mathbf{a} \in \Delta^{N-1}$, where a_i is the fraction of B delivered to candidate x_i ; the resource delivered is $B\mathbf{a}$. We model $\mathbf{a}$ as a density that evolves in time.

Definition 3.2 (Transport with a pump). Let μ_i be the *marginal value* of resource at candidate i (section 5 fixes $\mu_i = f_i$). The allocation obeys a discrete continuity equation with a conservative flux and a pump source:

$$\dot{a}_i = \underbrace{\sum_{j \sim i} J_{j \rightarrow i}}_{\text{conservative flux}} + \underbrace{\sigma_i(t)}_{\text{pump}}, \quad J_{j \rightarrow i} = -J_{i \rightarrow j}, \quad \sum_i \sigma_i(t) = 0, \quad (3)$$

where $j \sim i$ ranges over a coupling graph on the population, the flux is antisymmetric (resource leaving j arrives at i), and the pump σ redistributes without creating or destroying budget (e.g. a periodic re-seeding that injects exploration while drawing equally from the incumbents).

Theorem 3.3 (Budget conservation). *Under eq. (3), $\sum_i a_i$ is invariant; if $\mathbf{a}(0) \in \Delta^{N-1}$ then $\mathbf{a}(t) \in \Delta^{N-1}$ for all t , and the delivered resource $B \sum_i a_i = B$ is conserved.*

Proof. $\frac{d}{dt} \sum_i a_i = \sum_i \sum_{j \sim i} J_{j \rightarrow i} + \sum_i \sigma_i$. The first term is $\sum_{\{i,j\}} (J_{j \rightarrow i} + J_{i \rightarrow j}) = 0$ by antisymmetry; the second is 0 by the pump constraint. Hence $\sum_i a_i$ is constant; with initial sum 1 it stays 1. Nonnegativity is preserved provided the flux and pump vanish at the boundary $a_i = 0$ (no outflow from an empty candidate), which we assume of J, σ . $\square$

Remark 3.4 (The pump as rhythm and exploration). The pump σ is the theory’s “heart”. A natural choice is a slow oscillation $\sigma_i(t) = \epsilon \omega(t) (\pi_i - a_i)$ with $\sum_i \pi_i = 1$ a re-seeding distribution and ω periodic: it gently and periodically pulls the allocation toward π (exploration) without violating conservation, since $\sum_i \sigma_i = \epsilon \omega (\sum_i \pi_i - \sum_i a_i) = 0$ on the simplex. The oscillation prevents the allocation from collapsing onto a single incumbent and is the mechanism by which a system keeps “circulating” rather than fixating.

4 The Interoception Law

The field is not exogenous: it is set by the system’s reading of its own state. We give two interoceptive signals—a *coherence* and a *residual error*—and a homeostatic law driving the field toward a setpoint determined by them [2, 3, 6].

Definition 4.1 (Coherence and residual error). Assign to each candidate a *phase* $u(x_i) \in \mathbb{S}^{m-1}$ (a unit vector encoding the direction the candidate “points”, e.g. its dominant feature). The *coherence* of an allocation is the order parameter

$$R(\mathbf{a}) = \left\| \sum_i a_i u(x_i) \right\| \in [0, 1], \quad (4)$$

which is near 1 when the allocated mass points one way (the population agrees) and near 0 when it is dispersed [18, 22, 27]. The *residual error* is the expected intrinsic cost $\varepsilon(\mathbf{a}) = \sum_i a_i V(x_i)$.

Definition 4.2 (Interoceptive homeostasis). The field relaxes toward a setpoint $\mathbf{S}^*(R, \varepsilon)$ that depends on the interoceptive signals:

$$\dot{\mathbf{S}} = -\gamma(\mathbf{S} - \mathbf{S}^*(R(\mathbf{a}), \varepsilon(\mathbf{a}))), \quad \gamma > 0. \quad (5)$$

The map $\mathbf{S}^*$ encodes the system’s affective “policy”: for instance, $\mathbf{S}^*$ may raise the weight on exploratory affordances when coherence is high and error is not improving (escape a confident dead end), and raise the weight on consolidating affordances when coherence is low (gather the dispersed population).

Proposition 4.3 (Homeostatic stability). *For fixed interoceptive signals (hence fixed setpoint $\mathbf{S}^*$), the field dynamics eq. (5) is globally exponentially stable with rate γ : $\|\mathbf{S}(t) - \mathbf{S}^*\| \leq e^{-\gamma t} \|\mathbf{S}(0) - \mathbf{S}^*\|$.*

Proof. $\frac{d}{dt} \frac{1}{2} \|\mathbf{S} - \mathbf{S}^*\|^2 = \langle \mathbf{S} - \mathbf{S}^*, \dot{\mathbf{S}} \rangle = -\gamma \|\mathbf{S} - \mathbf{S}^*\|^2$, so $\|\mathbf{S} - \mathbf{S}^*\|^2$ decays as $e^{-2\gamma t}$ by Grönwall; take square roots [16]. $\square$

Remark 4.4 (Interoception in one line). Equations eq. (4)–eq. (5) are the formal content of “sentiment is interoception”: the field is a function of the system’s own coherence and error, i.e. of how the computation is going, and it feeds back to reshape (via eq. (1)) which candidates are stable. The field thus both *reports* and *regulates* the internal state, exactly as affect is held to do [4, 8].

5 The Economy

We now fix how, given the field, the budget is allocated. The principle is *maximum entropy under an expected-fitness constraint*: among all allocations achieving a given expected modulated benefit, choose the least committal [13]. This yields the Gibbs rule, which we show is the unique minimiser of a free energy and the stationary point of an entropy-regularised replicator dynamics, with the field appearing as the price of the budget.

Definition 5.1 (Fitness). The *fitness* of a candidate under field $\mathbf{S}$ is the negative modulated cost evaluated at its stable location: $f_i(\mathbf{S}) = -\Phi_{\mathbf{S}}(x_i) = -V(x_i) + \langle \mathbf{S}, \mathbf{g}(x_i) \rangle$. Higher fitness means lower modulated cost, hence greater claim on the budget.

Definition 5.2 (Free energy and the Gibbs allocation). For inverse temperature $\beta > 0$ define the *free energy*

$$\mathcal{F}_\beta(\mathbf{a}) = -\sum_i a_i f_i - \frac{1}{\beta} H(\mathbf{a}) = -\langle \mathbf{a}, \mathbf{f} \rangle - \frac{1}{\beta} H(\mathbf{a}), \quad \mathbf{a} \in \Delta^{N-1}, \quad (6)$$

and the *Gibbs allocation* $\mathbf{a}^\beta$ with $a_i^\beta = \frac{e^{\beta f_i}}{\sum_j e^{\beta f_j}}$.

Theorem 5.3 (The Gibbs allocation is the unique free-energy minimiser). $\mathcal{F}_\beta$ is strictly convex on Δ^{N-1} and attains its unique minimum at $\mathbf{a}^\beta$. Equivalently, $\mathbf{a}^\beta$ is the maximum-entropy allocation subject to a fixed expected fitness $\langle \mathbf{a}, \mathbf{f} \rangle$, and the field-induced terms $\langle \mathbf{S}, \mathbf{g}(x_i) \rangle$ enter f_i linearly, so the field acts as a vector of prices on the affordances.

Proof. On Δ^{N-1} , $-H$ is strictly convex (its Hessian is $\text{diag}(1/a_i) \succ 0$) and $-\langle \mathbf{a}, \mathbf{f} \rangle$ is linear, so $\mathcal{F}_\beta$ is strictly convex and has a unique minimiser [5]. Forming the Lagrangian $L = \mathcal{F}_\beta(\mathbf{a}) + \lambda(\sum_i a_i - 1)$ and setting $\partial L / \partial a_i = -f_i + \frac{1}{\beta}(\ln a_i + 1) + \lambda = 0$ gives $a_i \propto e^{\beta f_i}$; normalisation yields $\mathbf{a}^\beta$. The max-entropy characterisation is the standard duality: maximising $H(\mathbf{a})$ subject to $\langle \mathbf{a}, \mathbf{f} \rangle = c$ and $\sum a_i = 1$ has stationarity condition $\ln a_i = \beta f_i + \text{const}$ with β the multiplier on the fitness constraint [13]. The affordance prices are the components of $\mathbf{S}$ because $f_i = -V(x_i) + \sum_k S_k g_k(x_i)$. $\square$

Definition 5.4 (Entropy-regularised replicator dynamics). The allocation evolves by the *logit/replicator* flow

$$\dot{a}_i = a_i(f_i - \bar{f}) - \frac{1}{\beta} a_i(\ln a_i - \overline{\ln a}), \quad \bar{f} = \langle \mathbf{a}, f \rangle, \quad \overline{\ln a} = \langle \mathbf{a}, \ln \mathbf{a} \rangle, \quad (7)$$

the replicator dynamics [12, 19, 25] with an entropy-regularising drift. The unregularised flow ($\beta \rightarrow \infty$) is the classical replicator equation; the drift keeps the allocation in the interior.

Theorem 5.5 (Convergence of the allocation). *For fixed fitness f , the free energy $\mathcal{F}_\beta$ is a strict Lyapunov function for eq. (7) on $\text{int } \Delta^{N-1}$: $\dot{\mathcal{F}}_\beta \leq 0$ with equality only at $\mathbf{a}^\beta$. Hence every interior trajectory converges to the Gibbs allocation $\mathbf{a}^\beta$, which is globally asymptotically stable.*

Proof. Write $f_i^{\text{eff}} = f_i - \frac{1}{\beta} \ln a_i$, so eq. (7) is $\dot{a}_i = a_i(f_i^{\text{eff}} - \overline{f^{\text{eff}}})$ with $\overline{f^{\text{eff}}} = \langle \mathbf{a}, f^{\text{eff}} \rangle$, a replicator flow in the effective fitness. Note $f_i^{\text{eff}} = -\partial \mathcal{F}_\beta / \partial a_i + \text{const.}$ A standard computation gives $\dot{\mathcal{F}}_\beta = \sum_i \frac{\partial \mathcal{F}_\beta}{\partial a_i} \dot{a}_i = -(\overline{(f^{\text{eff}})^2} - (\overline{f^{\text{eff}}})^2) = -\text{Var}_{\mathbf{a}}(f^{\text{eff}}) \leq 0$, the replicator analogue of Fisher’s fundamental theorem [12]. Equality holds iff f^{eff} is constant across the support, i.e. $\ln a_i = \beta f_i + \text{const.}$, i.e. $\mathbf{a} = \mathbf{a}^\beta$. Strict convexity of $\mathcal{F}_\beta$ (theorem 5.3) makes $\mathbf{a}^\beta$ the unique minimum, so by LaSalle’s invariance principle [16] every interior trajectory converges to it. $\square$

Corollary 5.6 (Temperature controls commitment). *As $\beta \rightarrow \infty$ the Gibbs allocation concentrates on $\arg \max_i f_i$ (full commitment to the fittest candidate); as $\beta \rightarrow 0$ it tends to the uniform allocation (maximal exploration). The inverse temperature β is the system’s exploitation–exploration dial.*

Proof. Immediate from $a_i^\beta \propto e^{\beta f_i}$: the softmax tends to the indicator of the maximiser as $\beta \rightarrow \infty$ and to uniform as $\beta \rightarrow 0$ [7]. $\square$

6 Memory as the Field’s Path Integral

The field carries its own history. We define memory as the accumulated change of the field along the computation’s trajectory and record its elementary properties.

Definition 6.1 (Affective memory). The *affective memory* accumulated up to time t is the path integral of the field’s rate of change,

$$M(t) = \int_0^t \dot{\mathbf{S}}(\tau) \, d\tau = \mathbf{S}(t) - \mathbf{S}(0). \quad (8)$$

More finely, the *traversed memory* $\widetilde{M}(t) = \int_0^t \|\dot{\mathbf{S}}(\tau)\| \, d\tau$ records the total variation of the field along the path, distinguishing trajectories that reach the same field value by different routes.

Proposition 6.2 (Field–memory duality and path dependence). *M and $\mathbf{S}$ are a derivative–integral pair: $\dot{M} = \dot{\mathbf{S}}$ and $M(t) = \mathbf{S}(t) - \mathbf{S}(0)$. The reduced memory M depends only on the endpoints, but the traversed memory $\widetilde{M}$ is path-dependent: $\widetilde{M}(t) \geq \|M(t)\|$, with equality iff the field moves monotonically along a ray. Consequently two computations reaching the same field state through different affective histories are distinguished by $\widetilde{M}$.*

Proof. $\dot{M} = \dot{\mathbf{S}}$ and the endpoint formula are the fundamental theorem of calculus. The inequality is the triangle inequality for the path-length functional: $\widetilde{M} = \int_0^t \|\dot{\mathbf{S}}\| \geq \|\int_0^t \dot{\mathbf{S}}\| = \|M\|$, with equality iff $\dot{\mathbf{S}}$ keeps a fixed direction. $\square$

Remark 6.3. Definition 6.1 makes “memory is the trajectory, not the state” precise at the level of the field: what is remembered is *how the affect got here*, encoded as the integral of its motion, and the finer $\widetilde{M}$ separates histories that the coarse endpoint M conflates.

7 Metatheory of the Coupled System

The full system couples a fast allocation (eq. (7)) to a slow field (eq. (5)), with the field entering the allocation through the fitness $f_i(\mathbf{S})$ and the allocation entering the field through the interoceptive signals $R(\mathbf{a}), \varepsilon(\mathbf{a})$. We record its guarantees.

7.1 Boundedness

Proposition 7.1 (Boundedness). *If $\mathbf{S}^*$ is bounded on $[0, 1] \times \mathbb{R}_{\geq 0}$ and V is bounded on the (compact) region containing the population, then along any trajectory the allocation stays in Δ^{N-1} (theorem 3.3) and the field stays in a compact ball: $\limsup_t \|\mathbf{S}(t)\| \leq \sup \|\mathbf{S}^*\|$.*

Proof. The simplex is forward-invariant by theorem 3.3. For the field, with $B^* = \sup \|\mathbf{S}^*\|$, outside the ball $\|\mathbf{S}\| \leq B^* + \delta$ we have $\frac{d}{dt} \frac{1}{2} \|\mathbf{S}\|^2 = \langle \mathbf{S}, -\gamma(\mathbf{S} - \mathbf{S}^*) \rangle \leq -\gamma \|\mathbf{S}\|^2 + \gamma \|\mathbf{S}\| B^* < 0$, so the ball is attracting; hence $\limsup \|\mathbf{S}\| \leq B^*$ [16]. $\square$

7.2 The coherence phase transition

A central qualitative prediction is that the population’s coherence undergoes a sharp transition as the mutual coupling among candidates increases—the same mean-field transition that organises synchronization [18, 22]. We give a mean-field version.

Assumption 7.2 (Mean-field coupling). Suppose the field setpoint reinforces alignment: candidates are rewarded for agreeing with the population’s mean phase, so that the effective fitness contains a term $K R(\mathbf{a}) \cos(\theta_i - \psi)$ with coupling $K \geq 0$, where θ_i is candidate i ’s phase angle, ψ the mean-phase angle, and the candidates’ intrinsic phases are drawn from a symmetric, unimodal density $\rho(\omega)$ with spread σ_ω .

Theorem 7.3 (Coherence transition). *Under theorem 7.2 the fully incoherent allocation ($R = 0$) is the only stationary coherence for $K < K_c$ and becomes unstable for $K > K_c$, where the critical coupling is*

$$K_c = \frac{2}{\pi \rho(0)} \quad (\text{equivalently } K_c = c \sigma_\omega \text{ for a fixed shape constant } c). \quad (9)$$

For $K > K_c$ a coherent branch $R(K) > 0$ emerges continuously, $R(K) \sim \sqrt{K - K_c}$ near onset. Thus the population exhibits a turbulent regime ($R \approx 0$) below K_c and a coherent regime ($R > 0$) above it.

Proof sketch. This is the mean-field synchronization transition. Writing the self-consistency condition $R = KR \int_{-\pi/2}^{\pi/2} \cos^2 \theta \rho(KR \sin \theta) d\theta$ and expanding for small R gives $1 = K \frac{\pi}{2} \rho(0) + O(R^2)$, whence the onset $K_c = 2/(\pi \rho(0))$ and the square-root branch by the next order [18, 22]. Below K_c the only solution is $R = 0$; above it the incoherent solution is linearly unstable and the coherent branch is stable. $\square$

Remark 7.4 (Regimes as affective states). theorem 7.3 identifies the two coarse affective regimes with the two phases of a synchronization transition: dispersed/turbulent processing below the critical coupling, aligned/coherent processing above it. Because the coupling K is itself set by the field’s setpoint (theorem 7.2), the system can drive itself across the transition by interoceptive feedback—deliberately de-cohering to explore, or cohering to commit.

7.3 Stability of the coupled slow–fast system

Theorem 7.5 (Slow–fast stability). *Suppose the allocation is fast and the field slow: in eq. (7) replace time by t and in eq. (5) by ηt with $0 < \eta \ll 1$. Assume (i) for each fixed $\mathbf{S}$ the fast subsystem*

eq. (7) has the globally asymptotically stable equilibrium $\mathbf{a}^\beta(\mathbf{S})$ (theorem 5.5), depending smoothly on $\mathbf{S}$ (theorem 5.3); and (ii) the reduced slow dynamics $\dot{\mathbf{S}} = -\gamma(\mathbf{S} - \mathbf{S}^*(R(\mathbf{a}^\beta(\mathbf{S})), \varepsilon(\mathbf{a}^\beta(\mathbf{S}))))$ has a locally exponentially stable equilibrium $\mathbf{S}^\infty$. Then for η small enough the full coupled system has a locally exponentially stable equilibrium near $(\mathbf{a}^\beta(\mathbf{S}^\infty), \mathbf{S}^\infty)$.

Proof sketch. This is Tikhonov’s theorem for singularly perturbed systems [16, 17]. The fast allocation relaxes to the slow manifold $\mathbf{a} = \mathbf{a}^\beta(\mathbf{S})$ (an isolated, exponentially stable root of the fast equation by (i)); on that manifold the field evolves by the reduced dynamics, exponentially stable by (ii). The geometric singular-perturbation theorem then gives, for sufficiently small η , a persistent attracting invariant manifold $O(\eta)$ -close to the slow manifold and an equilibrium inheriting exponential stability from the reduced and boundary-layer systems. $\square$

Corollary 7.6 (The field as a slowly varying controller). *Under theorem 7.5 the allocation always tracks the current Gibbs allocation for the prevailing field, and the field migrates slowly toward the interoceptive setpoint. The system is therefore well-described, on the slow timescale, as a field-controlled allocator whose allocation is, at each instant, the optimal (free-energy-minimising) response to the field.*

8 Interface to a Host Runtime

The theory is designed to plug into a computational runtime as the component that biases which candidate computations proceed. We isolate the interface abstractly, so that the theory is independent of any particular host.

Definition 8.1 (Sentiment interface). A *host* supplies, at each instant: the candidate population C , the intrinsic costs $V(x_i)$, the affordances $\mathbf{g}(x_i)$, and the interoceptive signals it can measure (or the data from which R, ε are computed). The *sentiment module* returns: (i) the field $\mathbf{S}$ (and its memory M); and (ii) the allocation $\mathbf{a} = \mathbf{a}^\beta(\mathbf{S}) \in \Delta^{N-1}$, a priority distribution over candidates. The host uses $\mathbf{a}$ to schedule effort and consults M for the affective history.

Principle 8.2 (Sufficiency of the pair $(\mathbf{a}, M)$). For the purpose of deciding which computations proceed and with what priority, the pair $(\mathbf{a}, M)$ is a sufficient statistic of the field: the host need not see $\mathbf{S}$ itself, only the allocation it induces and the memory it has accumulated. This decouples the sentiment module from the host’s internal representation and makes the module replaceable.

Remark 8.3 (Relation to a scheduler). A host scheduler that draws the next unit of work according to $\mathbf{a}$ is, by theorems 5.5 and 7.6, sampling from the free-energy-minimising distribution for the current field; raising β sharpens it toward the fittest candidate (exploitation), lowering it flattens toward exploration (theorem 5.6). The field thus furnishes a principled, self-tuning, conserved priority that a host can consume without further modelling of affect.

9 A Worked Example

Consider three candidate lines of inquiry x_1, x_2, x_3 with intrinsic costs $V = (0.6, 0.4, 0.5)$ and two affordances, “novelty” g_1 and “consolidation” g_2 , with $\mathbf{g}(x_1) = (0.9, 0.1)$, $\mathbf{g}(x_2) = (0.2, 0.8)$, $\mathbf{g}(x_3) = (0.5, 0.5)$. Take inverse temperature $\beta = 4$.

A consolidating field. With $\mathbf{S} = (0.2, 1.0)$ (low novelty price, high consolidation price) the fitnesses are $f_i = -V_i + \langle \mathbf{S}, \mathbf{g}(x_i) \rangle$: $f_1 = -0.6 + (0.18 + 0.10) = -0.32$, $f_2 = -0.4 + (0.04 + 0.80) = 0.44$, $f_3 = -0.5 + (0.10 + 0.50) = 0.10$. The Gibbs allocation $a_i \propto e^{4f_i}$ is $\mathbf{a} \approx (0.06, 0.69, 0.25)$: the consolidating candidate x_2 dominates.

Interoceptive regime change. Suppose coherence is high ($R \approx 0.9$) but error has stopped improving. The setpoint $\mathbf{S}^*$ responds by raising the novelty price and lowering the consolidation price, say $\mathbf{S} \rightarrow (1.0, 0.2)$. Now $f_1 = -0.6 + (0.90 + 0.02) = 0.32$, $f_2 = -0.4 + (0.20 + 0.16) = -0.04$, $f_3 = -0.5 + (0.50 + 0.10) = 0.10$, and $\mathbf{a} \approx (0.55, 0.16, 0.29)$: the field has reallocated effort to the exploratory candidate x_1 *without any candidate being selected by fiat*—only the prices changed (theorem 2.3), the allocation tracked the new Gibbs minimiser (theorem 5.5), and the budget was conserved throughout (theorem 3.3). The memory $M = \mathbf{S}_{\text{after}} - \mathbf{S}_{\text{before}} = (0.8, -0.8)$ records the shift.

10 Related Work

The components of the theory each have a substantial pedigree, and locating them precisely clarifies the contribution.

Dynamical systems and bifurcation. The modulation law and its regime changes are standard equilibrium and bifurcation theory [11, 23]; the stability arguments use Lyapunov methods and LaSalle invariance [16].

Feedback and homeostasis. The interoception law is a homeostatic feedback in the cybernetic tradition [2, 3, 6, 26], with the coherence and error as the measured signals an observer would track [14].

Evolutionary dynamics and allocation. The economy is replicator dynamics with entropy regularisation [12, 19, 25]; its fixed point is the Gibbs/softmax allocation, the maximum-entropy distribution [7, 13] and the unique minimiser of a convex free energy [5]. The reading of the field as a price on affordances is the Lagrangian-duality view of allocation [1, 5, 15], and the whole apparatus is an instance of the “attention economy” for a resource-bounded processor [21].

Synchronization. The coherence order parameter and its phase transition are the mean-field theory of coupled oscillators [18, 22, 27].

Comparators. The free-energy formulation is formally adjacent to the free-energy principle for adaptive systems [10] and to the entropy-regularised objectives of reinforcement learning [24], but our free energy is over an *allocation* across candidate computations, not over actions or beliefs, and the controlling field is interoceptive. The interpretation of the field as affect follows the interoceptive theory of emotion [4, 8, 9].

The novelty is the synthesis: a single field that (a) modulates a variance landscape, (b) flows under a conservation law with a pump, (c) is driven by the system’s own coherence and error, and (d) clears a budget by a maximum-entropy economy—together with the theorems that make this a well-posed, convergent, conservative controller with a memory and a clean host interface.

11 Conclusion

We have given a self-contained mathematical theory of a sentiment field: a circulating, interoceptive, resource-allocating quantity that modulates which candidate computations stabilise and proceed. The four laws—modulation, circulation, interoception, economy—fit together into a coupled slow-fast dynamical system whose guarantees we proved: candidates are the minima of a field-tilted variance landscape and move smoothly with the field except at affective regime changes; the budget is conserved under transport with a pump; the field is driven homeostatically by the system’s own coherence and error; the allocation is the unique free-energy-minimising Gibbs distribution and is reached by an entropy-regularised replicator flow, with the field as the price of the budget;

coherence undergoes a sharp transition between turbulent and coherent regimes at a critical coupling; the coupled system is stable under timescale separation; and memory is the path integral of the field. The theory exposes a minimal interface—an allocation and a memory—through which a host runtime can consume sentiment as a self-tuning, conserved, accountable priority without modelling affect itself. It is intended both as a specification of the sentiment component of such a runtime and as a stand-alone account of why a field, rather than a scalar, is the right control variable for allocating the effort of an open-ended computation.

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